

The identity behind the sub-filter energy ledger: derivation and scope

G. Liu, 8 October 2026. Attachment to the letter on points 1–5.

This note derives, step by step, the identity used in the letter (its eq. 1) for the energy exchange between the waves and the part of the current that the Doppler term does not see, and states its scope. Every step says what it uses. Sections 1–7 are self-contained; §8 is a separate small-amplitude check that uses two results of Andrews & McIntyre.

1 Statement

Let the current be $\mathbf{u} = \mathbf{U} + \mathbf{u}'$, with $\mathbf{U} = \langle \mathbf{u} \rangle$ the filtered current and $\mathbf{u}'$ the rest, and let $\langle \cdot \rangle$ be a Reynolds average: linear, commuting with derivatives, idempotent, so $\langle \mathbf{u}' \rangle = 0$ and $\langle \mathbf{U}X \rangle = \mathbf{U}\langle X \rangle$. An ensemble average or a horizontal average is such an operator; a spatial filter of finite width is not (§9). Let $\nabla \cdot \mathbf{u}' = 0$, $\boldsymbol{\omega}' = \nabla \times \mathbf{u}'$, and let the Stokes drift $\mathbf{u}^S$ be deterministic and vary only on the filtered scales, so that $\langle \mathbf{u}^S X \rangle = \mathbf{u}^S \langle X \rangle$. Then

$$\rho \langle \mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') \rangle + \mathbf{u}^S \cdot \mathbf{F}^R = -\rho \mathbf{u}^S \cdot \nabla k_t, \quad \mathbf{F}^R \equiv -\rho \nabla \cdot \langle \mathbf{u}' \mathbf{u}' \rangle, \quad k_t \equiv \frac{1}{2} \langle |\mathbf{u}'|^2 \rangle. \quad (1)$$

2 Step 1: a pointwise identity, no assumption

For any vector field $\mathbf{b}$, in index notation with $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$,

$$[\mathbf{b} \times (\nabla \times \mathbf{b})]_i = \varepsilon_{ijk}b_j\varepsilon_{klm}\partial_l b_m = b_j\partial_i b_j - b_j\partial_j b_i = \partial_i \left(\frac{1}{2} |\mathbf{b}|^2 \right) - (\mathbf{b} \cdot \nabla) b_i.$$

With $\mathbf{b} = \mathbf{u}'$ and the sign reversed, $\boldsymbol{\omega}' \times \mathbf{u}' = (\mathbf{u}' \cdot \nabla) \mathbf{u}' - \nabla \frac{1}{2} |\mathbf{u}'|^2$. The scalar triple product is cyclic, $\mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') = \mathbf{u}^S \cdot (\boldsymbol{\omega}' \times \mathbf{u}')$, so

$$\mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') = \mathbf{u}^S \cdot [(\mathbf{u}' \cdot \nabla) \mathbf{u}' - \nabla \frac{1}{2} |\mathbf{u}'|^2]. \quad (2)$$

In words: the work of the Craik–Leibovich force on the fluid is the Stokes velocity dotted with the rotational part of the fluid’s own advective acceleration. This is the step (A2) of the barotropic calculation, applied to $\mathbf{u}'$.

3 Step 2: solenoidality

$(\mathbf{u}' \cdot \nabla) u'_i = u'_j \partial_j u'_i = \partial_j (u'_i u'_j) - u'_i \partial_j u'_j = \partial_j (u'_i u'_j)$, using $\nabla \cdot \mathbf{u}' = 0$. Hence

$$\mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') = u'_i \partial_j (u'_i u'_j) - u'_j \partial_j \left(\frac{1}{2} |\mathbf{u}'|^2 \right). \quad (3)$$

4 Step 3: the average

Apply $\langle \cdot \rangle$ to (3). Linearity and commutation with ∂_j give $\langle \partial_j (u'_i u'_j) \rangle = \partial_j \langle u'_i u'_j \rangle$; the assumption on $\mathbf{u}^S$ takes u'_i and u'_j through the average. Therefore

$$\langle \mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') \rangle = u'_i \partial_j \langle u'_i u'_j \rangle - \mathbf{u}^S \cdot \nabla k_t = -\frac{1}{\rho} \mathbf{u}^S \cdot \mathbf{F}^R - \mathbf{u}^S \cdot \nabla k_t,$$

and multiplying by ρ gives (??). □ Three facts were used, each once: the pointwise identity, $\nabla \cdot \mathbf{u}' = 0$, and the passage of $\mathbf{u}^S$ through the average.

5 Step 4: a second route, which shows where the Stokes production sits

Take the general identity $\mathbf{a} \times (\nabla \times \mathbf{b}) = \nabla(\mathbf{a} \cdot \mathbf{b}) - (\mathbf{a} \cdot \nabla) \mathbf{b} - (\mathbf{b} \cdot \nabla) \mathbf{a} - \mathbf{b} \times (\nabla \times \mathbf{a})$ with $\mathbf{a} = \mathbf{u}^S$, $\mathbf{b} = \mathbf{u}'$, and dot it with $\mathbf{u}'$; the last term drops because $\mathbf{u}' \cdot (\mathbf{u}' \times \cdot) = 0$:

$$\mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') = \underbrace{\mathbf{u}' \cdot \nabla(\mathbf{u}^S \cdot \mathbf{u}')}_{\text{I}} - \underbrace{\mathbf{u}' \cdot (\mathbf{u}^S \cdot \nabla) \mathbf{u}'}_{\text{II}} - \underbrace{\mathbf{u}' \cdot (\mathbf{u}' \cdot \nabla) \mathbf{u}^S}_{\text{III}}.$$

I = $\partial_i (u'_i u_j^S u'_j)$ by $\nabla \cdot \mathbf{u}' = 0$; expanding the product, I = $u_j^S \partial_i (u'_i u'_j) + u'_i u'_j \partial_i u_j^S$. II = $-(\mathbf{u}^S \cdot \nabla) \frac{1}{2} |\mathbf{u}'|^2$. III = $-u'_i u'_j \partial_j u_i^S$. The second piece of I and III add to $u'_i u'_j (\partial_i u_j^S - \partial_j u_i^S) = 0$, a symmetric tensor against an antisymmetric one, pointwise. What remains is (??) again. But before cancelling, average and read the three terms separately:

$$\rho \langle \mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') \rangle = \underbrace{-\rho \langle u'_i u'_j \rangle \partial_j u_i^S}_{\text{Stokes production}} + \underbrace{\partial_i [\rho \langle u'_i (\mathbf{u}^S \cdot \mathbf{u}') \rangle]}_{\text{transport}} - \underbrace{\rho \mathbf{u}^S \cdot \nabla k_t}_{\text{Stokes transport}}. \quad (4)$$

The first term is the Stokes production of the wave-averaged TKE equation, and (??) is the bookkeeping of that equation: production plus a transport divergence. Two bookkeepings of one quantity, not two contributions. Expanding the divergence makes this explicit:

$$\rho \langle \mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') \rangle = \underbrace{\rho \langle u'_i u'_j \rangle \partial_j u_i^S - \rho \langle u'_i u'_j \rangle \partial_j u_i^S + u_j^S \partial_i (\rho \langle u'_i u'_j \rangle)}_{=0 \text{ by symmetry}} - \rho \mathbf{u}^S \cdot \nabla k_t,$$

so the production term is cancelled pointwise by a piece of the transport divergence, and what survives is $-\mathbf{u}^S \cdot \mathbf{F}^R - \rho \mathbf{u}^S \cdot \nabla k_t$, which is (??). So (??) reads the Craik–Leibovich work as Reynolds force on $\mathbf{U}$ plus Stokes transport, and (??) reads it as Stokes production plus transport; the Stokes production and the $\mathbf{F}^R$ term are the same energy seen twice, and must not be added. Note that (??) has two transport terms, the divergence and the Stokes advection $-\rho \mathbf{u}^S \cdot \nabla k_t$; only if in addition $\nabla \cdot \mathbf{u}^S = 0$ does the second become a divergence too, $-\nabla \cdot (\rho k_t \mathbf{u}^S)$, so that

$$\rho \langle \mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') \rangle = -\rho \langle u'_i u'_j \rangle \partial_j u_i^S + \nabla \cdot [\rho \langle \mathbf{u}' (\mathbf{u}^S \cdot \mathbf{u}') \rangle - \rho k_t \mathbf{u}^S],$$

the local Stokes production plus one transport divergence. Without $\nabla \cdot \mathbf{u}^S = 0$, (??) is the form to use. Locally the two readings assign the energy to different places; in the volume integral they agree when the transport has no flux through the boundaries. None of this algebra is new. The TKE budget of the wave-averaged equations with the Stokes production term is McWilliams, Sullivan & Moeng (1997, their eq. 5.1); the rewriting of the vortex force and Bernoulli head as Stokes advection plus a Stokes shear force, of which only the latter exchanges energy with the waves, is Suzuki & Fox-Kemper (2016), and their terms are I, II and III above: the head and the Stokes advection are the transports, the Stokes shear force work is the production; and the energy compartments of the wave-averaged equations, including the cross energy $\mathbf{U} \cdot \mathbf{u}^S$, are set out by Czeschel & Eden (2023), who remark that a physical interpretation of that cross energy is lacking. What this note adds is the reading of the surviving term in Step 5: $-\mathbf{u}^S \cdot \mathbf{F}^R$ is the Doppler tendency of the filtered current, the cross energy is the Doppler energy of the wave, and so the sub-filter current needs no Doppler account of its own.

6 Step 5: what the three terms of (??) are

Left, first term. In the Craik–Leibovich equations the wave acts on the current through the vortex force $\rho \mathbf{u}^S \times \boldsymbol{\omega}$ and a Bernoulli head $-\rho \nabla K$. On the sub-filter current the vortex force does the work $\rho \langle \mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') \rangle$ at all of its scales; the head does $-\rho \langle \mathbf{u}' \cdot \nabla K' \rangle = -\rho \nabla \cdot \langle \mathbf{u}' K' \rangle$, a pure flux with $\nabla \cdot \mathbf{u}' = 0$, zero through boundaries where $w' = 0$. So the first term is the rate at which the waves feed sub-filter kinetic energy.

$\mathbf{u}^S \cdot \mathbf{F}^R$. $\mathbf{F}^R$ is the force that the sub-filter stress exerts on the filtered current in the filtered momentum equation, $\rho \partial_t \mathbf{U} = \dots + \mathbf{F}^R$. The wave model sees $\mathbf{U}$ only through the Doppler shift $\boldsymbol{\omega} = \boldsymbol{\sigma} + \mathbf{k} \cdot \tilde{\mathbf{U}}$, $\tilde{\mathbf{U}} = \int_{-\infty}^0 2ke^{2kz} \mathbf{U} dz$ (Stewart & Joy 1974), so the Doppler energy of the wave, $\rho \mathbf{k} \cdot \tilde{\mathbf{U}} N$, changes at the rate $\rho N \mathbf{k} \cdot \partial_t \tilde{\mathbf{U}}$ at fixed N ; the term $\rho \mathbf{k} \cdot \tilde{\mathbf{U}} \dot{N}$ is not part of this exchange, since the exchange leaves N unchanged (§8). Now $N \mathbf{k} 2ke^{2kz} = \mathbf{u}^S$ as vectors: with $N = ga^2/2\sigma$ and $\sigma^2 = gk$, $N \mathbf{k} \cdot 2ke^{2kz} = ga^2 k^2 e^{2kz} / \sigma = a^2 \sigma k e^{2kz} = \mathbf{u}^S$. (N is the action per unit area and unit density, so $\rho \sigma N = E_{\text{int}}$ is the usual E/σ times σ ; readers who put ρ inside N should read ρN for N throughout.) Hence the part of the Doppler tendency due to $\mathbf{F}^R$ is

$$\rho N \mathbf{k} \cdot \partial_t \tilde{\mathbf{U}} \Big|_{\mathbf{F}^R} = \int \rho \mathbf{u}^S \cdot \frac{1}{\rho} \mathbf{F}^R dz = \int \mathbf{u}^S \cdot \mathbf{F}^R dz.$$

So $\mathbf{u}^S \cdot \mathbf{F}^R$ is the rate at which the sub-filter current changes the wave’s Doppler energy, through $\mathbf{U}$, per unit volume. The step rests on one structural fact, that the Doppler weight of the wave is proportional to its Stokes drift profile; in finite depth the weight is $2k \cosh 2k(z+h) / \sinh 2kh$ (Kirby & Chen 1989) and the Stokes drift is again proportional to it, so the identification is not limited to deep water.

Right. $-\rho \mathbf{u}^S \cdot \nabla k_t$ is the advection of sub-filter kinetic energy by the Stokes drift; with $\nabla \cdot \mathbf{u}^S = 0$ it is $-\nabla \cdot (\rho \mathbf{u}^S k_t)$, a transport. It vanishes identically for a horizontal average, where ∇k_t is vertical and $\mathbf{u}^S$ horizontal, and it integrates to zero over a closed domain.

Reading. Integrated over a closed domain, with $\nabla \cdot \mathbf{u}^S = 0$ and no flux of the Bernoulli head through the boundaries, (??) says: the energy that the Craik–Leibovich force puts into the sub-filter current balances the energy that the sub-filter current takes out of the wave’s Doppler account by pushing on the filtered current. Locally the two differ by the transport terms of (??), which (??) keeps explicit. The wave side and the current side are computed from different equations, the action equation with $\boldsymbol{\omega} = \boldsymbol{\sigma} + \mathbf{k} \cdot \tilde{\mathbf{U}}$ and the current momentum equation, and agree by (??). At the resolved level the same statement is $C = -\rho \mathbf{p} \cdot \partial_t \mathbf{u}$ of the barotropic calculation, with $\mathbf{U}$ in place of $\mathbf{u}'$; the only difference is that the resolved current enters $\boldsymbol{\omega}$ directly, while the sub-filter current enters it through $\mathbf{F}^R$ on $\mathbf{U}$. This exchange changes $\boldsymbol{\omega}$ and not N ; §8 checks this for the monochromatic wave.

7 Step 6: the horizontally homogeneous check

With $\mathbf{u}^S = u^S(z) \hat{\mathbf{x}}$ and horizontal averaging, $F_x^R = -\rho \partial_z \langle u' w' \rangle = \partial_z \bar{\tau}$ with $\bar{\tau} = -\rho \langle u' w' \rangle$ the mean shear stress carried by the sub-filter motion, and the right side vanishes. Write $\tau_s \equiv \bar{\tau}(0)$ for its surface value and $\mathcal{W}_S \equiv \tau_s u^S(0)$, $\mathcal{P}_S \equiv \int \bar{\tau} \partial_z u^S dz$, as in the letter. Then

$$\rho \langle \mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') \rangle = -u^S \partial_z \bar{\tau}, \quad \int_{-\infty}^0 \rho \langle \mathbf{u}' \cdot (\mathbf{u}^S \times \boldsymbol{\omega}') \rangle dz = -[u^S \bar{\tau}]_{-\infty}^0 + \int \bar{\tau} \partial_z u^S dz = \mathcal{P}_S - \mathcal{W}_S,$$

and $\int \mathbf{u}^S \cdot \mathbf{F}^R dz = \int u^S \partial_z \bar{\tau} dz = \mathcal{W}_S - \mathcal{P}_S$, the Doppler tendency $\rho N d\boldsymbol{\omega}/dt$ of a sheared column due to the mean stress, in which $\bar{\tau}$ is the total mean stress continued to the surface, $\bar{\tau}(0) = \tau_s$ the wind stress. That Doppler tendency is the depth integral of (??) when the stress at the surface is carried by the sub-filter motion itself. If instead $w' = 0$ at the surface and the surface stress is carried there by viscosity or by the SGS part, then $\bar{\tau}(0) = 0$ for the part covered by (??), the boundary term vanishes, and $\mathcal{W}_S$ enters the Doppler equation through the viscous or SGS part of $\partial_t \mathbf{U}$ instead, with the same total $\mathcal{W}_S - \mathcal{P}_S$. In this horizontally homogeneous geometry the Stokes advection term vanishes and the only transport is the divergence of (??), $\partial_z (\rho \langle w' u' \rangle u^S)$; its depth integral is $-\mathcal{W}_S$, which is the difference between the Craik–Leibovich work $\mathcal{P}_S - \mathcal{W}_S$ and the Stokes production $\mathcal{P}_S$.

8 Check of the action projection for a monochromatic deep-water wave

The claim at the end of Step 5 is that the energy the sub-filter current takes from the waves through $\mathbf{F}^R$ comes out of the Doppler account and not out of the action. This is not part of the Reynolds-average identity (??); it is a separate small-amplitude calculation, done here for the linear monochromatic deep-water wave to $O(a^2)$, with two ingredients from Andrews & McIntyre (1978a,b). First, the forcing term of the GLM pseudomomentum equation: a body force $\mathbf{X}$ per unit mass changes $p_i = -\langle \xi_{j,i} u_j^\ell \rangle$ at the rate $-\langle \xi_{j,i} X_j^\ell \rangle$, where X^ℓ is the Lagrangian disturbance of the force, the force felt by a parcel

minus its mean. Second, for a linear plane wave the depth-integrated pseudomomentum is k times the action, $\rho \int p_x dz = \rho k N$, and $E_{\text{int}} = \rho \sigma N$, so the action changes as $\rho \sigma \dot{N} = c \rho \int \partial_t p_x dz$ with $c = \sigma/k$ the phase speed; both relations hold at the order kept here. Take the sub-filter stress carried with the fluid with its components frozen, which is the leading-order behaviour of turbulence whose own rates are slow compared with σ (the rapid-distortion limit; whatever the turbulence does beyond being carried is, by definition, $\tau^\times$), so that the parcel of label $\bar{z}$ keeps its mean stress $\bar{\tau}(\bar{z})$ and the Eulerian stress is $\bar{\tau}(\bar{z}(x, z, t))$, with $\bar{z}(x, z, t)$ the inverse of the particle map $x_i = \bar{x}_i + \xi_i$. Then $\partial_z \tau_{xz} = \bar{\tau}'(\bar{z}) \partial_z \bar{z}$ and $\partial_x \tau_{zx} = \bar{\tau}'(\bar{z}) \partial_x \bar{z}$, with $\partial \bar{x}_j / \partial x_i$ the inverse of the deformation gradient $\partial x_i / \partial \bar{x}_j = \delta_{ij} + \xi_{i,j}$; to first order in the slope, $\partial_z \bar{z} = 1 - \xi_{z,z}$ and $\partial_x \bar{z} = -\xi_{z,x}$, so the force on the parcel is $\bar{\tau}'(1 - \xi_{z,z})$ in x and $-\bar{\tau}' \xi_{z,x}$ in z , and $X^\ell = -\bar{\tau}'(\bar{z}) (\xi_{z,z}, \xi_{z,x}) / \rho$. The derivative $\bar{\tau}'$ is taken at the parcel's own label; nothing is expanded about a fixed level. With the deep-water displacements $\xi_x = -ae^{kz} \sin \theta$, $\xi_z = ae^{kz} \cos \theta$, $\theta = kx - \sigma t$,

$$-\rho \langle X_x^\ell \xi_{z,x} \rangle = k \bar{\tau}' \langle \xi_z \xi_{z,x} \rangle = -\frac{1}{2} k^2 a^2 e^{2kz} \bar{\tau}', \quad -\rho \langle X_z^\ell \xi_{z,x} \rangle = \bar{\tau}' \langle \xi_z^2 \rangle = +\frac{1}{2} k^2 a^2 e^{2kz} \bar{\tau}',$$

and the two cancel. The carried normal stresses $R_{xx}(\bar{z})$ and $R_{zz}(\bar{z})$ give forces $-R'_{xx} \xi_{z,x}$ in x and $-\partial_z (\xi_z R'_{zz})$ in z , in quadrature with $\xi_{x,x}$ and $\xi_{z,x}$ respectively, and project to zero. So to $O(a^2)$ the carried mean stress has zero action projection, and its whole effect on the wave is the Doppler change $\int \mathbf{u}^S \cdot \mathbf{F}^R dz$ of Step 5. Only $ak \ll 1$ was used; no condition on the vertical scale of $\bar{\tau}$. What does project on the action is the wave-coherent part of the stress beyond pure displacement, $\tau^\times$, and the surface traction; both are outside (??).

9 Scope: what was used, and what remains outside

1. $\nabla \cdot \mathbf{u}' = 0$ (Step 2). Boussinesq or incompressible fluctuations.
2. $\mathbf{u}^S$ passes through the average (Step 3). Exact for an ensemble average and for a horizontal average under a plane wave. For a spatial filter of width ℓ it requires $\mathbf{u}^S$ to vary on scales large compared with ℓ , i.e. the wave envelope wider than the filter.
3. $\langle \mathbf{u}' \mathbf{u}' \rangle$ is the stress of the filtered momentum equation (Step 5). Two situations have to be kept apart.

Horizontal average of a horizontally periodic LES. The horizontal average is a Reynolds operator, so (??) holds exactly for the resolved fields, with $\mathbf{u}'$ the resolved departure from the horizontal mean and $\langle \mathbf{u}' \mathbf{u}' \rangle$ the resolved Reynolds stress. The SGS stress enters the Doppler side through $\partial_t \mathbf{U}$ and the current side through its modelled Stokes production, $\tau_{xz}^{\text{SGS}} \partial_z \mathbf{u}^S$; the two agree only as well as the SGS model does. This is the split of point 2 of the letter.

A Doppler filter of finite width. If $\mathbf{U}$ is instead defined by a spatial filter at the scale $\sim 1/k$ (point 4 of the letter), with $\mathbf{U} = \langle \mathbf{u} \rangle$ and $\mathbf{u}' = \mathbf{u} - \mathbf{U}$, the stress in the filtered equation is $\tau_{ij}^{\text{SGS}} = \langle u_i u_j \rangle - U_i U_j = L_{ij} + C_{ij} + \langle u'_i u'_j \rangle$, with the Leonard term $L_{ij} = \langle U_i U_j \rangle - U_i U_j$ and the cross term $C_{ij} = \langle U_i u'_j \rangle + \langle u'_i U_j \rangle$ (Leonard 1974; the sum is an identity for any linear filter). The wave side sees τ^{SGS} through $\partial_t \mathbf{U}$; the identity has $\langle u'_i u'_j \rangle$; the mismatch is $-\rho u_i^S \partial_j (L_{ij} + C_{ij})$. For a Reynolds average $\langle U_i U_j \rangle = U_i U_j$ and $\langle U_i u'_j \rangle = U_i \langle u'_j \rangle = 0$, so both vanish. For a filter of finite width both are in general non-zero, a sharp spectral cut-off included: the product of two resolved fields has content beyond the cut-off, so $L = -(1 - \langle \cdot \rangle)(U_i U_j) \neq 0$, and $C \neq 0$ even though $\langle \mathbf{u}' \rangle = 0$.

4. The Craik–Leibovich form of the wave forcing (Step 5): vortex force and Bernoulli head, the head doing no net work on $\mathbf{u}'$.
5. The Doppler current is $\tilde{\mathbf{U}}$ with weight $2ke^{2kz}$ and the filtered current obeys $\rho \partial_t \mathbf{U} \ni \mathbf{F}^R$ (Step 5). Only the part of $\partial_t \mathbf{U}$ due to $\mathbf{F}^R$ is used; other forces on $\mathbf{U}$ have their own entries.

In one sentence: (??) is an exact Reynolds-average identity, not a generic finite-width-filter identity; for an LES filter the Leonard and cross contributions have to be retained. Outside the identity, by construction: any action of the sub-filter current on the waves that does not pass through $\mathbf{F}^R$ on $\mathbf{U}$. These are the resonant scattering by eddies with $|q| \leq 2k$, with the recoil as its current-side partner, and the wave-coherent stress of the eddies, $\Pi^\times$. They are the other two entries of the ledger in the letter, and are not closed here.

Checks

Every step above, including §8, is computed symbolically in `verify_identity.py`: the pointwise identities on a general solenoidal field and a general $\mathbf{u}^S(x, y, z)$, the averaging step on a two-member ensemble, the Doppler weight, the depth integral, the two projections of §8 and the vanishing of the normal-stress projections, and, numerically, the decomposition of item 3 of §9 for a Gaussian filter and for a sharp cut-off.

References. Andrews & McIntyre 1978a *JFM* 89, 609; 1978b *JFM* 89, 647. Czeschel & Eden 2023, arXiv:2311.13354. Kirby & Chen 1989 *JGR* 94, 1013. Leonard 1974 *Adv. Geophys.* 18A, 237. McWilliams, Sullivan & Moeng 1997 *JFM* 334, 1. Stewart & Joy 1974 *Deep-Sea Res.* 21, 1039. Suzuki & Fox-Kemper 2016 *JGR Oceans* 121, 3579.